Technology Stack

Date	17 September 2026
Team ID	SWTID-2026-3976
Project Name	AI FAQ Assistant API (AnswerBot)
Maximum Marks	2 Marks

Define Your Technology Stack:

The technologies below were chosen for a secure, scalable and maintainable backend.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	None (headless API). Consumed by Postman, ThunderClient, web, mobile or admin clients	The project is a backend-first REST service so any client can use it.
2	Backend / Server-Side	Node.js 18+, Express.js, dotenv, cors	Fast, non-blocking JavaScript runtime; Express gives simple routing and middleware.
3	Database / Data Storage	MongoDB (Atlas) with Mongoose ODM	Flexible document storage for FAQs and users; Mongoose adds schemas and validation.
4	Authentication and Security	jsonwebtoken (JWT), bcrypt, validation middleware	Stateless sessions, strong password hashing and route guarding.
5	Cloud / Hosting / Deployment	Local development server (port 5001) with MongoDB Atlas cloud database	Simple to run and demo; can be deployed later to Render, Railway or AWS.
6	Version Control & CI/CD	Git and GitHub	Team collaboration and change history.
7	Third-Party APIs / Other Tools	Google Gemini API (@google/genai); Postman / ThunderClient; VS Code	Enables AI answer and FAQ generation; API testing; development environment.